

Reflective Lantern - Innovation Report

New Project Built

Name: Suite-Cast

Branch: <https://github.com/atharvadevne123/reflective-lantern/tree/innovation/suite-cast>

Pull Request: <https://github.com/atharvadevne123/reflective-lantern/pull/22>

Date: 2026-07-15

Inspired by: Hospitality revenue-management gap in portfolio (HN/GitHub trending blocked by egress policy; topic chosen via duplicate-gate elimination)

What it does: Hotel booking demand forecasting and dynamic pricing API using an XGBoost-LightGBM ensemble with 18 engineered features, KS-test drift monitoring, SQLAlchemy prediction logging, and champion/challenger retraining.

Note: session GitHub access is scoped to reflective-lantern, so standalone repo creation, tag pushes, and release creation were blocked. Project shipped on branch innovation/suite-cast with draft PR #22. A one-click publisher workflow (bootstrap-suite-cast.yml, Actions tab, needs a repo-scope PAT) creates the standalone Suite-Cast repo, tag, release, and wheel asset. Root-level CI (suite-cast-ci.yml) runs lint, tests, and wheel build on the branch and PR. Wheel verified locally: suite_cast-1.0.0-py3-none-any.whl.

Tech Stack Coverage: 100%

[X] Python

[X] FastAPI/Flask

[X] XGBoost/LightGBM/RF

[X] sklearn Pipeline

[X] Model monitoring

[X] Docker+compose

[X] SQLAlchemy

[X] pytest suite

[X] .env.example

[X] GitHub Actions CI

Optional picks (4+): time-series/seasonality features, drift detection, automated retraining (Airflow champion/challenger DAG), 5-fold CV + AUC-ROC, Alembic migrations, rate limiting, correlation-ID tracing.

Project Stats

Files created: 37

Tests written: 100 (all passing)

Commits: 62

Lint: ruff check + format clean